Md Imran Khan

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PROFESSIONAL SUMMARY

Ph.D. optical physicist and CD-SEM metrology engineer with a rare combination of first-principles wave-optics modeling, optical system design, and production-deployed imaging analytics. Built full-wave electromagnetic simulators for plasmonic nano-structures and the custom optical hardware to validate them; ~4 years of semiconductor metrology experience qualifying imaging tools for NPI and deploying Python-based IQA, anomaly detection, and SPC trend monitoring across an 8+ tool fleet. Actively building computer vision and deep learning systems (U-Net segmentation, Grad-CAM, image quality assessment) and targeting metrology, imaging systems, sensor analytics, and CV / ML roles.

TECHNICAL SKILLS

Metrology & Process Control: CD-SEM, SEM, optical metrology, gauge capability (GR&R / Cp / Cpk), SPC, tool-to-tool matching, measurement uncertainty, DOE design and execution, measurement recipe development

Image Quality & Vision Systems: Image quality analysis (CNR, sharpness, FFT, GLCM, SSIM, PSNR, MTF), imaging sensor characterization, anomaly detection, semantic segmentation (U-Net), OpenCV, PyTorch

Signal Processing & Analytics: DSP, FFT, filtering, time-series analysis, multivariate statistical modeling, anomaly detection pipelines, sensor data analytics, noise analysis & SNR

Programming & Data: Python (NumPy, pandas, Matplotlib, scikit-learn, OpenCV, PyTorch, scikit-image, FastAPI), MATLAB, SQL/MySQL, LabVIEW, JMP, Git, Jupyter, Origin, Zemax

Optics & Physical Modeling: Optical system design & alignment, light–matter interaction simulation (MFS, Foldy-Lax), illumination geometry (bright-field, dark-field, structured light), polarization, TRPL system design, UV–IR broadband modeling, confocal microscopy

Fabrication & Characterization: Clean room, sputtering, photolithography, nano-fabrication, DLS, DSC, integrating sphere spectroscopy

EXPERIENCE

Independent AI / Computer Vision Engineer — August 2025 – Present

Self-Directed | Portland, OR

- Full-time focused transition into applied computer vision, deep learning, and ML engineering roles in imaging systems, computational imaging, and scientific / medical imaging.
- Designed, built, and deployed end-to-end ML systems and a production full-stack web application spanning semiconductor metrology, microstructure segmentation, and medical imaging — see Technical Projects.
- Structured self-study across deep learning, classical computer vision (Szeliski), digital signal processing, and statistical pattern recognition (Bishop's PRML).

Metrology Process Engineer – CD-SEM — November 2021 – July 2025

Intel Corporation | Hillsboro, OR

- Developed and deployed a Python-based image quality analysis pipeline applying multiple IQA algorithms (sharpness, CNR, FFT high-frequency energy, GLCM) to characterize SEM imaging sensor performance and resolve a recurring measurement drift issue that had been managed reactively for over 3–4 years — transitioning a team of 5 engineers from reactive troubleshooting to proactive, data-driven maintenance.
- Designed and deployed an anomaly detection system monitoring high-dimensional sensor data streams across a fleet of 8 CD-SEM tools; enabled proactive identification of imaging system degradation before production impact.
- Managed fleet-wide gauge capability and measurement stability; applied SPC, automated calibration logic, and GR&R studies to sustain Cp/Cpk targets and minimize measurement drift across advanced process nodes.
- Automated multivariate statistical pipelines for tool-to-tool matching and measurement precision evaluation; reduced data processing time by 80% (5 hrs to 1 hr) while improving fleet-wide consistency and reporting accuracy.
- Independently developed and deployed the first long-term SPC trend monitoring pipeline in the workgroup using SQLPathfinder and Python — enabling fleet-wide longitudinal visibility into metrology performance.
- Defined measurement sampling strategies and gauge capability specifications to qualify new process layers and product lines following tool installation or process integration changes.
- Performed structured root cause investigations on electron-optical drift and imaging anomalies; coordinated corrective actions across hardware, software, and process integration teams.

- Authored technical documentation and training materials; trained technicians on tool maintenance, calibration procedures, and SPC interpretation to ensure production readiness.

Graduate Research Scholar – Optics & Computational Modeling August 2016 – October 2021

University of California, Merced | Merced, CA

- Developed a complete physics-based simulation platform for modeling light–matter interactions in 3D plasmonic meta-structures using hybrid full-wave methods (MFS and Foldy-Lax theory); computed far-field scattering, extinction spectra, and broadband optical response across UV–IR wavelengths.
- Designed and commissioned a custom high-stability TRPL optical measurement system; defined specifications, aligned optical components, developed calibration protocols, and validated performance against simulation predictions.
- Independently executed end-to-end experimental workflow: nano-fabricated core-shell plasmonic structures, characterized using SEM, DLS, integrating sphere spectroscopy, and automated optical measurements, and validated results against simulation with strong agreement.
- Developed automated data acquisition and analysis pipelines in Python and MATLAB; published peer-reviewed research in JOSA A and Optics Express and presented at APS conferences.

Research Associate 2013 – 2014

Universal Instruments Corporation | Kirkwood, NY

- Characterized thermomechanical properties of 92.5Pb-5Sn-2.5Ag solder alloys using DSC and SEM; co-authored an IMAPS industry publication on material composition and process performance.

TECHNICAL PROJECTS

SEM Image Quality Assessment (IQA) Web App | *FastAPI, Streamlit, Docker, OpenCV | sem-iqm-tool.streamlit.app* *Oct 2025 – Feb 2026*

- Designed and deployed a full-stack web application for SEM imaging sensor diagnostics with a Dockerized FastAPI backend (hosted on Render) and a Streamlit frontend (hosted on Streamlit Community Cloud) across a two-repo architecture; implements full-reference (SSIM, PSNR) and no-reference (BRISQUE, Laplacian sharpness, FFT high-frequency energy, variance-based focus) image quality metrics with interactive Plotly visualizations — applicable to metrology tool qualification, imaging sensor characterization, and inline inspection.

SEM Semantic Segmentation Pipeline | *PyTorch, Python, OpenCV, PIL* *March 2026*

- Built a multi-class semantic segmentation pipeline (U-Net) for SEM microstructure images to automatically segment 5 material phases; implemented class-weighted CE + Dice loss and per-class IoU/Dice evaluation — demonstrating end-to-end ML model development for scientific image analysis.

Chest X-Ray Pneumonia Classification | *PyTorch, scikit-learn, Python* *November 2025*

- End-to-end pneumonia detection pipeline on chest X-ray images comparing classical machine learning (PCA + SVM, ROC-AUC ~0.92) and deep learning approaches (CNN from scratch and transfer learning with ResNet-18 / 34 / 50, ROC-AUC ~0.95) for multi-class classification (Normal, Viral, Bacterial); applied Grad-CAM to visualize model attention on lung opacities — demonstrating end-to-end ML methodology spanning classical baselines, deep learning, and interpretability for medical imaging.

EDUCATION

Ph.D., Physics – Optics & Computational Modeling October 2021

University of California, Merced

NSF-CREST Graduate Scholar — Center for Cellular and Biomolecular Machines (CCBM)

Graduate Scholar Mentor — mentored undergraduate researchers on experimental methods and graduate school preparation

M.S., Physics May 2016

Binghamton University (SUNY)

SELECTED PUBLICATIONS

Khan, M.I., et al., "Scattering by nanoplasmonic mesoscale assemblies," JOSA A 42, 1244–1253 (2025).

Khan, M.I., et al., "Modeling broadband cloaking using 3D nano-assembled plasmonic meta-structures," Optics Express (2020).

Schoeller, H., Anselm, M., Khan, I., & Cotts, E., "Effect of Sn Component Surface Finish on 92.5Pb-5Sn-2.5Ag," HITEC Conf. Proc., Vol. 2014, pp. 364–371.